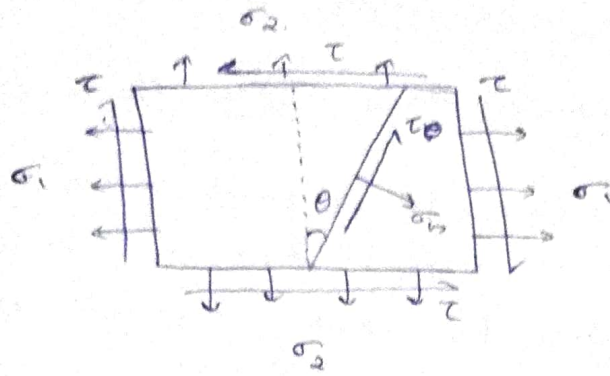


22-1-25



$$\sigma_n = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta + \tau \sin 2\theta$$

$$\tau_\theta = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta - \tau \cos 2\theta$$

To find principle plane, $\tau_\theta = 0$ i.e., shear = 0.

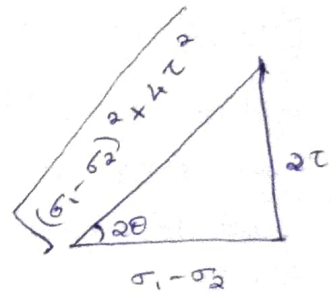
$$\tau_\theta = 0 \Rightarrow \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta = \tau \cos 2\theta$$

$$\frac{\sin 2\theta}{\cos 2\theta} = \frac{2\tau}{\sigma_1 - \sigma_2}$$

$$\tan 2\theta = \frac{2\tau}{\sigma_1 - \sigma_2}$$

$$\sin 2\theta = \frac{\pm 2\tau}{\sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}}$$

$$\cos 2\theta = \frac{\pm (\sigma_1 - \sigma_2)}{\sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}}$$



$$\sigma_{n1} = \frac{\sigma_1 + \sigma_2}{2} + \frac{1}{2} \sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}$$

$$\sigma_{n2} = \frac{\sigma_1 + \sigma_2}{2} - \frac{1}{2} \sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}$$

Max stress condition.

Min "

$$\tau_{max} = \frac{\sigma_{n1} - \sigma_{n2}}{2} = \frac{1}{2} \sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}$$

$$\sigma_2 = 0$$

$$\sigma_{n1} = \frac{\sigma_1}{2} + \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2}$$

$$\sigma_{n2} = \frac{\sigma_1}{2} - \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2}$$

$$\tau_{max} = \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2}$$

If stress is in only 1-direction.

Theories of failure under static load.

Max principle or normal stress theory (Rankine theory).

Failure or yielding occurs at a point in a member when the maximum principle or normal stress in a bi-axial stress system ~~reaches~~ reaches the limiting strength of the material in a simple tension stress.

For ductile,

$$\sigma_{n1} = \frac{\sigma_{ye}}{F.S.} \approx \frac{\text{Yield point}}{\text{Factor of safety}}$$

$$F.S. = \frac{\text{limiting load}}{\text{working load.}}$$

For brittle,

$$\sigma_{n1} = \frac{\sigma_u}{F.S.}$$

Max shear stress theory (Guests or Trescas theory).

Failure or yielding occurs at a point in a member when the max shear stress in a bi-axial stress system reaches a value equal to shear stress at yield point in a simple tension test.

$$\tau_{max} = \frac{\tau_{yl}}{F.S.}$$

$$\tau_{yl} = \frac{\sigma_{yl}}{2}$$

$$\therefore \tau_{max} = \frac{\sigma_{yl}}{2 F.S.}$$

Max. Principle strain theory (Saint Venant's theory)

Failure or yielding occurs at a point in a member when the max principle strain in a bi-axial stress system reaches the limiting value of strain (strain at yield point).

Max strain in a bi-axial stress system $\Rightarrow$

$$E_{max} = \frac{\sigma_{n1}}{E} - \frac{\sigma_{n2}}{mE}$$

Theory

$$E_{\max} = \frac{\sigma_{n1}}{E} - \frac{\sigma_{n2}}{mE} = \frac{\sigma_{yl}}{EFS}$$

$E \Rightarrow$ modulus of elasticity

$$\frac{\sigma_{n1} - \frac{\sigma_{n2}}{m}}{E} = \frac{\sigma_{yl}}{FS}$$

Max strain energy theory (Heigh's theory).

Failure or yielding occurs at a point in a member whether strain energy per unit volume in a bi-axial stress-system reaches the limiting strain energy (strain energy at yield point) per unit volume

Generally, strain energy, $U = \frac{\sigma^2}{2E}$

strain energy per unit volume in a biaxial systems;

$$U_1 = \frac{1}{2E} \left[\sigma_{n1}^2 + \sigma_{n2}^2 - \frac{2\sigma_{n1}\sigma_{n2}}{m} \right]$$

Theory $\Rightarrow$

$$\frac{1}{2E} \left[\sigma_{n1}^2 + \sigma_{n2}^2 - \frac{2\sigma_{n1}\sigma_{n2}}{m} \right] = \frac{1}{2E} \left[\frac{\sigma_{yl}}{FS} \right]^2$$

$$\boxed{\sigma_{n1}^2 + \sigma_{n2}^2 - \frac{2\sigma_{n1}\sigma_{n2}}{m} = \left[\frac{\sigma_{yl}}{FS} \right]^2}$$

Max distortion energy theory (Hencky & Von. Mises theory)

The failure or yielding occurs at a point in a member when the distortion strain energy (shear strain energy) per unit volume in a bi-axial stress system reaches the limiting distortion energy (distortion energy at yield point) per unit volume.

29-1-25
1. The load on a ball consists of an axial pull of 10kN together with a transverse shear force of 5kN. Find the dia of ball required according to (i) max principal stress theory.

- ii). max shear stress theory (iii). max principal strain theory,
 (iv). max strain energy theory & (v). max distortion energy theory

Given the permissible tensile stress at elastic limit = 100 MPa
 & Poisson's ratio = 0.3

Ans. $P_1 = 10 \text{ kN}$.

Shear force, $\tau = 5 \text{ kN}$.

Permissible stress = $\sigma_p = 100 \text{ MPa}$.

Poisson's ratio, = 0.3

$d = ?$

$$\sigma_1 = \frac{P_1}{A} = \frac{10 \times 10^3}{\frac{\pi}{4} d^2} = \frac{12.73}{d^2} \text{ kN/mm}^2.$$

$$\tau = \frac{5}{\frac{\pi}{4} d^2} = \frac{6.36}{d^2} \text{ kN/mm}^2.$$

$$\sigma_{n1} = \frac{\sigma_1 + \sigma_2}{2} + \frac{1}{2} \sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}$$

$$= \frac{\sigma_1}{2} + \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2}$$

$$\therefore \sigma_{n1} = \frac{12.73}{2d^2} + \frac{1}{2} \sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4\left(\frac{6.36}{d^2}\right)^2}$$

$$= \frac{12.73}{2d^2} + \frac{1}{2} \cdot \sqrt{\frac{162.05}{d^4} + \frac{161.79}{d^4}}$$

$$\sigma_{n1} = \frac{12.73}{2d^2} + \frac{1}{2} \sqrt{\frac{323.85}{d^4}}$$

$$= \frac{12.73}{2d^2} + \frac{1}{2} \cdot \frac{17.99}{d^2}$$

$$= \frac{12.73}{2d^2} + \frac{8.99}{d^2} = \frac{21.73}{d^2}$$

$$= \frac{15.35}{d^2}$$

$$\begin{aligned}
 i). \quad \tau_{max} &= \frac{1}{2} \sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2} \\
 &= \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2} \\
 &= \frac{1}{2} \sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4 \cdot \left(\frac{6.36}{d^2}\right)^2} \\
 &= \frac{1}{2} \sqrt{\frac{162.05}{d^4} + \frac{52.77 + 161.79}{d^4}} \\
 &= \frac{1}{2} \sqrt{\frac{323.84}{d^4}} \\
 &= \frac{1}{2} \cdot \frac{17.99}{d^2} = 8.99 \\
 &\approx \frac{9}{d^2} \text{ kN} = \frac{9000}{d^2} \text{ N}
 \end{aligned}$$

$$\sigma_{n, \text{ permissible}} = \sigma_p = \frac{\sigma_{yl}}{F.S}$$

$$\frac{15360}{d^2} = 100 \Rightarrow d^2 = \frac{15360}{100} = 153.6$$

$$\therefore d = \underline{\underline{12.4 \text{ mm}}}$$

$$ii). \quad \tau_{max} = \tau_p = \frac{\sigma_p}{2}$$

$$\frac{9000}{d^2} = \frac{100}{2}$$

$$d^2 = \frac{9000 \times 2}{100} = 180 \Rightarrow d = \underline{\underline{13.4 \text{ mm}}}$$

$$iii). \quad \frac{\sigma_{n1} - \sigma_{n2}}{m} = \sigma_p$$

$$\begin{aligned}
 \sigma_{n2} &= \frac{\sigma_1}{2} = \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2} \\
 &= \frac{12.73}{2d^2} - \frac{1}{2} \sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4 \cdot \left(\frac{6.36}{d^2}\right)^2}
 \end{aligned}$$

$$= \frac{6.36}{d^2} - \frac{1}{2} \sqrt{\frac{162.05}{d^4} + \frac{161.79}{d^4}}$$

$$= \frac{6.36}{d^2} - \frac{1}{2} \sqrt{\frac{323.84}{d^4}}$$

$$= \frac{6.36}{d^2} - \frac{1}{2} \cdot \frac{17.99}{d^2}$$

$$= \frac{6.36}{d^2} - \frac{8.99}{d^2}$$

$$= \frac{-2.63}{d^2} \text{ kN/mm}^2 = \frac{-2630}{d^2} \text{ N/mm}^2.$$

$$\frac{15360}{d^2} - \left[\frac{-2630}{d^2} \right] 0.3 = 100.$$

$$\frac{16149}{d^2} = 100 \Rightarrow d^2 = 161.49$$

$$\therefore d = \underline{\underline{12.71 \text{ mm}}}$$

$$\text{iv). } \sigma_{n1}^2 + \sigma_{n2}^2 - \frac{2\sigma_{n1}\sigma_{n2}}{m} = \sigma_p^2.$$

$$\left(\frac{15360}{d^2} \right)^2 + \left(\frac{2630}{d^2} \right)^2 - 2 \times \frac{15360}{d^2} \times \frac{2630}{d^2} \times 0.3 = 100^2$$

$$\frac{235929600}{d^4} + \frac{6916900}{d^4} + \frac{24238080}{d^4} = 100^2.$$

$$d = \underline{\underline{12.78 \text{ mm}}}$$

$$\sigma_{n1}^2 + \sigma_{n2}^2 - \frac{2\sigma_{n1}\sigma_{n2}}{m} = \sigma_p^2$$

$$\left(\frac{15360}{d^2} \right)^2 + \left(\frac{2630}{d^2} \right)^2 - 2 \times \frac{15360}{d^2} \times \frac{2630}{d^2} = 100^2.$$

$$\frac{235729600}{d^4} + \frac{6916900}{d^4} + \frac{80793600}{d^4} = 100.$$

$$\therefore d = \underline{\underline{13.41 \text{ mm}}}$$

2. A cylindrical shaft made up of steel of yield strength 700 MPa is subjected to static loads consisting of bending moment 10 kNm and a torsional moment of 30 kNm. Determine the dia of shaft using 2 different theories of failure. Assuming a factor of safety of 2 take young's modulus = 210 GPa & Poisson's ratio 0.25.

Ans. $E = 210 \text{ GPa}, \quad M = 10 \text{ kNm}, \quad T = 30 \text{ kNm}.$

Poisson's ratio = 0.25 = μ .

$\sigma_{yl} = 700 \text{ MPa}$

FOS = 2.

$$\sigma_{ni} = \frac{\sigma_{yl}}{\text{FOS}} = \frac{700 \times 10^6}{2} = \underline{\underline{350 \text{ MPa}}}.$$

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = y \cdot \frac{M}{I} = \frac{M}{Z}$$

$$Z = \frac{\pi}{32} d^3 = \underline{\underline{0.098 d^3}}.$$

$$\sigma = \frac{10 \times 10^3 \times 32 \times 10^3}{\pi d^3} = \frac{101.85 \text{ MN/mm}^2}{d^3}$$

$$\tau = \frac{16 T}{\pi d^3} = \frac{16 \times 30 \times 10^3 \times 10^3}{\pi d^3} = \frac{152.788 \text{ MN/mm}^2}{d^3}$$

$$= \underline{\underline{\frac{152.8 \text{ MN/mm}^2}{d^3}}}$$

$$\sigma_{n1} = \frac{\sigma_1}{2} + \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2}$$

$$= \frac{101.85 \times 10^6}{2d^3} + \frac{1}{2} \sqrt{\left(\frac{101.85 \times 10^6}{d^3}\right)^2 + 4 \cdot \left(\frac{152.8 \times 10^6}{d^3}\right)^2}$$

$$= \frac{50925000}{d^3} + \frac{1}{2} \sqrt{\frac{1.037 \times 10^{16}}{d^6} + \frac{9.339 \times 10^{16}}{d^6}}$$

$$= \frac{50925000}{d^3} + \frac{1}{2} \cdot \frac{3.2 \times 10^8}{d^3}$$

$$= \frac{50925000}{d^3} + \frac{1}{2} \cdot \frac{644235981.6}{d^3} = \frac{1.61 \times 10^8}{d^3}$$

$$= \frac{2.11 \times 10^8}{d^3} = \frac{211 \times 10^6}{d^3}$$

$$\sigma_{n2} = \frac{\sigma_1}{2} - \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2}$$

$$= \frac{101.85 \times 10^6}{2d^3} - \frac{1}{2} \sqrt{\left(\frac{101.85 \times 10^6}{d^3}\right)^2 + 4 \cdot \left(\frac{152.8 \times 10^6}{d^3}\right)^2}$$

$$= \frac{50925000}{d^3} - \frac{1}{2} \sqrt{\frac{1.037 \times 10^{16}}{d^6} + \frac{9.339 \times 10^{16}}{d^6}}$$

$$= \frac{50925000}{d^3} - \frac{1.61 \times 10^8}{d^3}$$

$$= \frac{-110.13 \times 10^6}{d^3}$$

$$\tau_{max} = \frac{1}{2} \sqrt{\sigma_1^2 + 4\tau^2} = \frac{\sigma_{n1} - \sigma_{n2}}{2}$$

$$= \left[\frac{211 \times 10^6}{d^3} - \frac{-110.13 \times 10^6}{d^3} \right] \frac{1}{2}$$

$$= \frac{161 \times 10^6}{d^3}$$

$$\tau_{max} = \frac{\sigma_{yl}}{2FS}$$

$$\frac{161 \times 10^6}{d^3} = \frac{700}{2 \times 2}$$

$$\therefore d = \underline{\underline{97.26 \text{ mm}}}$$

$$\sigma_{n1}^2 + \sigma_{n2}^2 - \frac{2\sigma_{n1}\sigma_{n2}}{m} = \left(\frac{\sigma_{yl}}{FS} \right)^2$$

$$\left(\frac{211.91 \times 10^6}{d^3} \right)^2 + \left(\frac{110.13 \times 10^6}{d^3} \right)^2 - 2 \times \frac{211.91 \times 10^6}{d^3} \times \frac{110.13 \times 10^6}{d^3} \times 0.25$$

$$= \left(\frac{700}{2} \right)^2 = 122500$$

$$\frac{4.49 \times 10^{16}}{d^6} + \frac{1.21 \times 10^{16}}{d^6} + \frac{1.166 \times 10^{16}}{d^6} = 122500$$

$$\frac{6.866 \times 10^{16}}{d^6} = 122500$$

$$\therefore d = \underline{\underline{90.8 \text{ mm}}}$$

3. A mild steel shaft of 50mm dia is subjected to a BM of 2000 Nm & a torque, T. If the yield point of the steel in tension is 200MPa, find the max value of this torque without causing yielding of shaft according to max principal stress, max shear stress & max distortion strain energy theory of yielding.

1. A double rivetted lap joint is made b/w 15mm thick plates. The rivet dia & pitch are 25 & 75mm respectively. If the ultimate stress are 400 MPa in tension, 320 MPa in shear & 640 MPa in crushing, find the min force per pitch which will rupture the joint. If the above joint is subjected to a load such that the factor of safety is 4, find out the actual stresses developed in the plates & the rivets.

Ans. $t = 15\text{mm}$, $d = 25\text{mm}$, $P = 75\text{mm}$, $\sigma_t = 400\text{MPa}$,
 $\tau = 320\text{MPa}$, $\sigma_c = 640\text{MPa}$, $FS = 4$,

$$P_t, P_s, P_c \dots$$

$$P_t = (P-d) t \sigma_t$$

$$= (75-25) 15 \times 400 = \underline{\underline{3000000 \text{ N}}}$$

$$P_s = n \frac{\pi}{4} d^2 \tau = 2 \frac{\pi}{4} 25^2 \times 320 = \underline{\underline{314159.2 \text{ N}}}$$

$$P_c = n d t \sigma_c$$

$$= 2 \times 25 \times 15 \times 640 = \underline{\underline{4800000 \text{ N}}}$$

Min force required for rupture $\Rightarrow \underline{\underline{3000000 \text{ N}}}$.

$$\text{Permissible force} = \frac{3000000}{4} = \underline{\underline{750000 \text{ N}}}$$

$$P_t = (P-d) t \sigma_t$$

$$75000 = (75-25) 15 \sigma_{ta} \Rightarrow \sigma_{ta} = \underline{\underline{100 \text{ MPa}}}$$

$$P_s = n \frac{\pi}{4} d^2 \tau \Rightarrow 2 \tau$$

$$75000 = n \frac{\pi}{4} d^2 \tau_a$$

$$= 2 \frac{\pi}{4} \cdot 25^2 \tau_a \Rightarrow \tau_a = \underline{\underline{76.4 \text{ MPa}}}$$

$$P_c = n d t \sigma_c$$

$$75000 = 2 \times 25 \times 15 \times \sigma_{ca} \Rightarrow \sigma_{ca} = 100 \text{ MPa}.$$

2. Find η of following riveted joints

- i). Single riveted lap joint of 6mm plates with 20mm dia, rivets having a pitch of 50mm.
- ii). Double riveted lap joint of 6mm plates with 20mm dia rivets having a pitch of 65mm.

Permissible tensile stress in plate = 100 MPa.

" shearing stress in rivets = 90 MPa



" crushing stress in rivets = 180 MPa.

